

# Moorthy Chetan

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## Summary

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Second-year B.Tech student with hands-on experience building full-stack web applications using React, Node.js, and PostgreSQL. Interested in creating performant, secure, and user-focused web products.

## Skills

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**Languages & Frameworks:** JavaScript, React, Next.js, Node.js, Express, SQL, Python

**Tools & Technologies:** Socket.IO, PostgreSQL, MongoDB, Git, REST APIs, Web Crypto API

## Projects

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**End-to-End Encrypted Real-Time Chat App (PrivChat)** *Next.js, Node.js, Socket.IO, PostgreSQL, Web Crypto API*

- Architected a zero-knowledge messaging system where encryption and decryption happen entirely in the browser using AES-GCM (per-message) and RSA-OAEP (key wrapping), ensuring the server never has access to plaintext.
- Generated RSA-4096 keypairs in the browser and persisted them in IndexedDB; public keys are synced to the server while private keys never leave the device.
- Implemented real-time messaging with Socket.IO and Google OAuth authentication via `passport-google-oauth20`.
- Designed a PostgreSQL schema to store only ciphertext, IVs, and per-recipient encrypted message keys — making the backend fully zero-knowledge.

**Real-Time Collaborative Study Platform (FocusSync)** *React, Node.js, Express, Socket.IO, MongoDB*

- Built a server-controlled synchronized focus timer shared across all room participants in real time using Socket.IO, with late-join sync so users entering mid-session receive the correct remaining time.
- Implemented distraction detection using the Page Visibility API, tracking tab-switch events and computing a per-user focus score.
- Developed a multi-page analytics dashboard with session history, leaderboard (top rooms by focus time), and date-grouped heatmaps via REST API endpoints.
- Structured the app with a React + React Router frontend and a Node.js/Express backend with Mongoose models for session persistence.

## Education

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**Vellore Institute of Technology, Vellore**  
B.Tech in Information Technology

2024 – 2028  
CGPA: 8.4